microorganisms play a major role. This is because of their high enzymatic activity and their ubiquity.

Most macro elemental components of living organisms like Carbon, Nitrogen, Sulphur, Oxygen, Phosphorus and Hydrogen are cycled most intensely and other elements like Copper, Chromium, Iron, Zinc, Nickel are cycled less intensely.

11.1.1 Carbon Cycle

Carbon is a macro element present in all living cells. In microorganisms, they are present in all macromolecules like cell wall, cytoplasmic membrane, proteins and nucleic acids.

Reservoirs of Carbon:

Reservoirs are the storage places of nutrients that are present in nature. They store nutrients in large amounts for longer periods of time.

Atmospheric CO_2, dissolved carbon in oceans and freshwater, organic matter are actively cycled carbon reservoirs. Sediments and fossil fuels like coal, petroleum and natural gas are slowly cycled carbon reservoirs. Carbon is cycled between these reservoirs by the biochemical activities of micro organisms and other living things (Figure 11.1).

The different stages or processes involved in carbon cycle are

1. Photosynthesis
2. Decomposition
3. Methanogenesis

1. Photosynthesis

It is a process where atmospheric CO_2 is converted to organic carbon $(CH_2O)n$. This is carried out by higher plants, photosynthetic bacteria, cyanobacteria and algae using radiant energy from

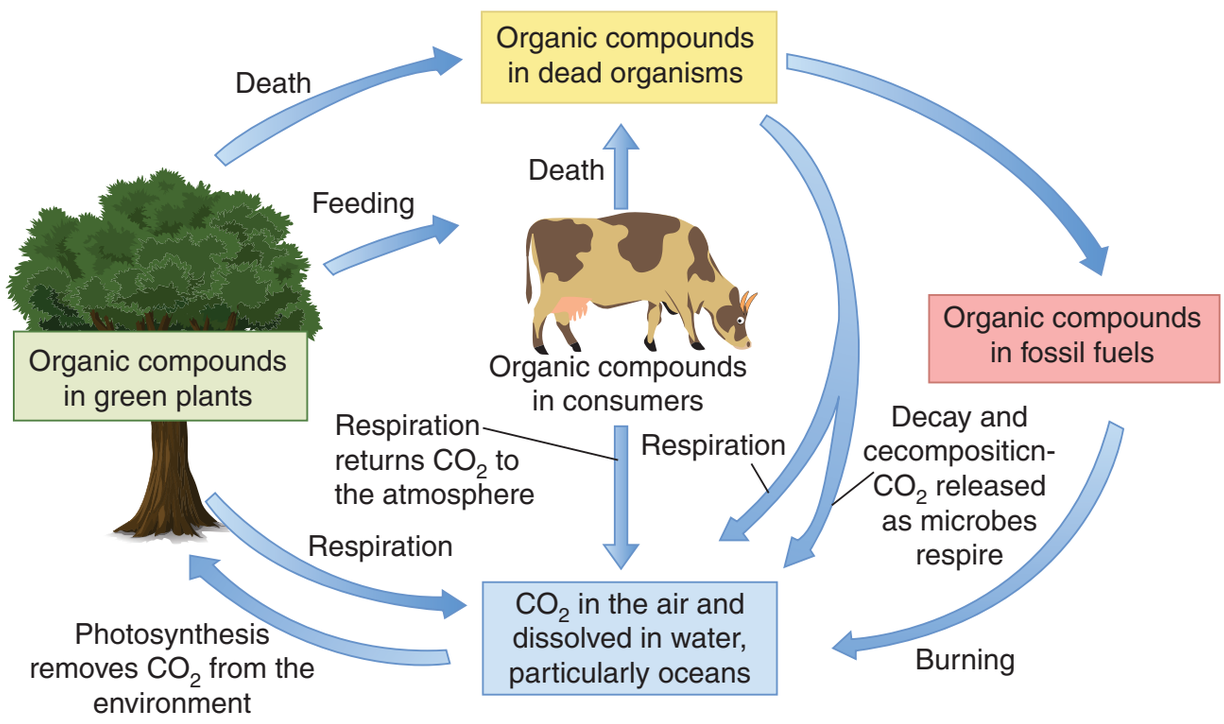


Figure 11.1: A simplified diagram of Carbon cycle is as follows

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